1. I labeled some of the statements labeled other as Realtor Network and Media Outreach. It could be a little of a stretch for these, but they are there just for reference.
2. For the machine learning classifier, I'm using TF-IDF to turn the words into numbers. And logistic regression to train the model. The idea is both from Cursor and the discussion group. What I see is that with so little data and such a wide range of categories, with some categories having only 1-2 responses, it can't ever get any accurate result, especially on the less common ones.
3. I used a training size of 20 and testing size of 10 given that we have 30 responses. And I tested the algorithm by running it 10 times each time randomizing the train and testing datasets.
4.

Class	**Precision**	**Recall**	**F1**	**Specificity**	**One-vs-rest AUC**
buyer	0.579	0.983	0.722	0.090	0.796
seller	N/A	0.000	N/A	1.000	0.500
renter	N/A	0.000	N/A	1.000	0.749
media outreach	N/A	0.000	N/A	1.000	0.294

realtor network	N/A	0.000	N/A	1.000	0.500
other	0.000	0.000	0.000	0.943	0.671

Macro metric	**Mean across 10 repeats**
Macro precision	0.434
Macro recall	0.235
Macro F1	0.579
Macro specificity	0.839
Macro AUC	0.631

Actual \ Predicted	**buyer**	**seller**	**renter**	**media outreach**	**realtor network**	**other**
buyer	55	0	0	0	0	1

seller	6	0	0	0	0	0
renter	11	0	0	0	0	0
media outreach	6	0	0	0	0	4
realtor network	2	0	0	0	0	0
other	15	0	0	0	0	0

The above are data we are getting from the python notebook.

The problem of the algorithm is that it's not really predicting that much. Almost all predictions are buyers because that's appearing the most in the training data. So the safest bet for the algorithm is to just guess "buyer" on the vast majority if not all predictions to get the highest accuracy result. If we do not divide the other category, it could have done better.